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Application of the Green–Ampt model for infiltration into layered soils

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SUMMARY

Using the Green–Ampt model for layered soils, a theoretical method is proposed to estimate the layers' hydraulic parameters based on infiltration test data. To evaluate the proposed method and to investigate the effect of the layers' permeability on infiltration, constant head infiltration tests were conducted on six different two-layered soil columns. When the upper layer was more permeable than the lower layer, piston flow was observed in both layers, and the air-filled volume fraction of the lower layer was nearly equal to the observed values in homogenous soils. In this situation, the layers' hydraulic parameters can be estimated from the proposed method. When the lower layer was more permeable than the upper layer, piston flow with a large air-filled volume fraction was observed in the lower layer. When the lower layer was significantly more permeable with an approximately zero wetting front matric head, preferential flows or instability in the wetting front was observed in the lower layer. In two-layered soils with a more permeable layer below a low permeable layer, the hydraulic conductivity of the lower layer cannot be estimated based on infiltration test data.

1. Introduction

Infiltration is an important component of the hydrological cycle by which surface runoff and groundwater recharge can be linked. Infiltration models can be separated into three categories: physical, approximately physical, and empirical. Although empirical models such as that proposed by Kostiakov (1932) are popular because of their simplicity and capability of fitting well to measured infiltration data, they contain parameters that are difficult to predict based on soil's physical characteristics. Green and Ampt (1911) directly applied Darcy's law and proposed an approximately physical model for water infiltration from a ponded surface into a deep homogeneous soil with a uniform initial water content. This model assumes a constant hydraulic conductivity and a constant water content in the wetting zone, and a constant (negative) head at the wetting front (Bouwer, 1969). To apply the Green–Ampt infiltration model, soil surface must be ponded. When the rainfall rate is greater than the soil infiltration rate capacity, the soil surface will be ponded. Additionally, in ponded irrigation systems and groundwater recharge basins, the soil surface is typically ponded. To simulate infiltration from groundwater recharge basins, the Green–Ampt infiltration model was proposed by Bouwer (2002). Combining Darcy's law with the mass conservation law, Richards (1931) derived a partial differential equation for a one-dimensional vertical flow in an unsaturated soil that was physically based. Infiltration is a dependent boundary condition of the soil water movement equations. To accurately simulate infiltration using Richards' equation, the soil water movement must be accurately simulated (Smith and Woolhiser, 1971). Richards' equation is strongly nonlinear, and except for special cases, a general analytical solution is not possible for this equation. Consequently, numerical methods such as the finite difference and finite element methods have been used to solve Richards' equation (Arampatzis et al., 2001). The numerical solution of Richards' equation requires an iterative implicit technique with a fine spatial discretization, which results in a tedious solving process (Damodhara Rao et al., 2006). Compared to Richards' equation, the Green–Ampt model is simpler and can be directly used to describe infiltration.

Due to the natural stratification caused by soil formation, fauna and management in farmlands, the soil profile is frequently layered (Fig. 1). The hydraulic parameters and therefore permeability of each soil layer is usually different from other soil layers; therefore, infiltration into layered soils is more complex compared to homogeneous soils.

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List of symbols

Af_1	air-filled volume fraction of the 1st layer
Af_2	air-filled volume fraction of the 2nd layer
d	diameter range of the soil particles (L)
H	constant water depth above the soil surface (L)
I	cumulative infiltration depth (L)
I_1	total infiltrated water to soil at $t = t_1$ (L)
I_2	total infiltrated water to soil at $t = t_2$ (L)
I_{i-1}	total infiltrated water to soil at $t = t_{i-1}$ (L)
K_1	hydraulic conductivity of the 1st layer (L/T)
K_2	hydraulic conductivity of the 2nd layer (L/T)
K_{i-1}	hydraulic conductivity of the $(i-1)$th layer (L/T)
K_i	hydraulic conductivity of the ith layer (L/T)
K_s	saturated hydraulic conductivity (L/T)
$\overline{K}$	equivalent hydraulic conductivity from the ground surface to the wetting front (L/T)
i	hydraulic gradient (L/L)
l	an empirical parameter (–)
L	thickness of the wetting zone (L)
L_1	thickness of the 1st layer (L)
L_2	thickness of the 2nd layer (L)
L_{i-1}	thickness of the $(i-1)$th layer (L)
L_i	thickness of the ith layer (L)
n	number of layers; and an empirical parameter
t	time (T)
t_1	wetting front arrival time at the bottom of the 1st layer (T)
t_2	wetting front arrival time at the bottom of the 2nd layer (T)
t_{i-1}	wetting front arrival time at the bottom of the $(i-1)$th layer (T)
α	an empirical parameter (1/L)
ρ_s	specific density (M/L^3)
Ψ_{f1}	wetting front matric head of the 1st layer (L)
Ψ_{f2}	wetting front matric head of the 2nd layer (L)
Ψ_{fi}	wetting front matric head of the ith layer (L)
Φ_1	porosity of the 1st layer
Φ_2	porosity of the 2nd layer
θ_i	initial water content
θ_r	residual water content
θ_s	saturated water content
$\Delta\theta_1$	wetness increment of the 1st layer
$\Delta\theta_2$	wetness increment of the 2nd layer
$\Delta\theta_i$	wetness increment of the ith layer

Generally, when the wetting front of the ponded infiltration reaches the interface of two soil layers, the infiltration rate at the wetting front arrival time to the interface is dominated by the sorptivity, which is the capacity of the soil medium to absorb or desorb water by capillarity (Philip, 1957), of the lower soil layer. For a system with a coarse soil layer above a fine soil layer, when the wetting front of the ponded infiltration reaches the interface between the two layers, the infiltration rate may marginally increase due to the greater attraction between the finer soil layer and the water (Miller and Gardner, 1962). Conversely, the greater sorptivity of the finer soil layer tends to increase the infiltration rate. As the wetting front advances into the finer soil layer, the water must be transmitted through the finer pores, which produce an increased resistance to flow (Latifi et al., 1994). The resistance to flow may be so great that the flow rate is significantly reduced. Consequently, due to the significant reduction in the infiltration rate, a step change in the infiltration rate data can be observed after wetting front arrival time to the interface (Miller and Gardner, 1962). When the wetting front of the ponded infiltration moves from a fine soil layer to a coarse soil layer, the infiltration rate is believed to decrease at the wetting front arrival time to the interface of the two layers, because the attraction is reduced between the larger pores in the coarser soil and the water (Latifi et al., 1994). Conversely, when the wetting front arrives at the interface, a smaller sorptivity of coarser soil layer produces a step reduction in the infiltration rate. In this situation, the lower coarse soil layer will not be saturated (Chu and Mariño, 2005). The movement of a one-dimensional wetting front in homogeneous soil is known to be stable; however, in a layered soil whose upper layer is finer and less permeable than the coarser lower layer, the wetting front becomes unstable and breaks into narrow wetting columns or fingers (Hill and Parlange, 1972).

Few models that were based on the Green–Ampt approach were proposed to simulate infiltration into layered soils. Warrick (1991) developed two approximations for infiltration into a stratified soil profile. The first (i.e., the square front solution) assumes that the soil profile is fully wetted up to the wetting front and satisfies the initial condition below the wetting front. The second solution (i.e., the homogeneous analog solution) assumes that the stratified soil profile is analogous to a homogeneous soil profile. In this model, the two-term Philip (1969) algebraic expression was used to simulate the surface infiltration; no comparison with the field or laboratory infiltration data was performed in this study. Corradini et al. (2000) proposed an analytical/conceptual model to simulate the local infiltration into a two-layered soil under complex rainfall patterns. The model was tested by comparison with the numerical solution of Richards' equation and was shown to simulate the entire cycle of infiltration–redistribution–reinfiltra tion accurately. To determine the ponding condition and infiltration into layered soils under an unsteady rainfall, a numerical algorithm based on the Green–Ampt model was used by Chu and Mariño (2005). Comparisons with the other infiltration models (i.e., the modified Green–Ampt infiltration model and the fully numerical model) and field measurements were conducted, and good agreements were found. Using the Green–Ampt approach, Liu et al. (2008) derived an infiltration model for layered soils with a non-uniform initial water content and unsteady infiltration. Three infiltration experiments were conducted to validate the model, and comparisons indicated that the model was in good agreement with another Green–Ampt-based model and the measured data. To simulate the water infiltration into a five-layered laboratory soil column, Ma et al. (2010, 2011) proposed a modified version of the Green–Ampt model. This model provided satisfactory simulation results and adequately described the infiltration process in a laboratory soil column. A conceptual model of local infiltration into a two-layered soil profile with a significantly more permeable upper layer was proposed by Corradini et al. (2011a); this model was a reformulation of the conceptual model proposed by Corradini et al. (2000) that involved coupled ordinary differential equations that described the infiltration and soil water content distribution in a two-layered soil under any rainfall pattern. The conceptual model was then extended by Corradini et al. (2011b) from local infiltration to the field-scale infiltration.

To estimate the deep percolation and therefore groundwater recharge from the layered soils and to estimate the surface runoff on layered soils, infiltration into layered soils must be investigated. The existing models for simulating infiltration into layered soils are usually complex and computationally difficult to apply at large

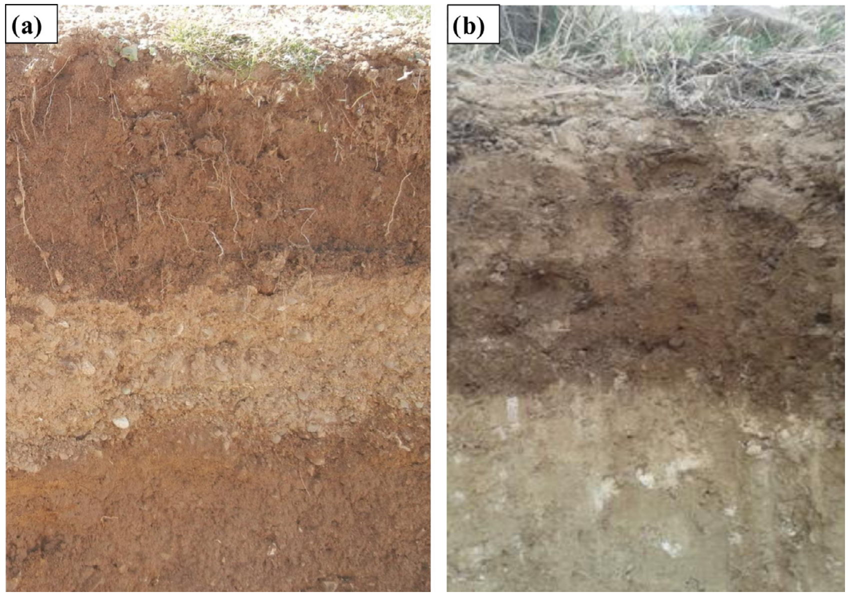

Fig. 1. Images of layered soils profile (a) due to the natural stratification, a high permeable layer (sandy soil layer) is placed between the two low permeable layers and (b) due to the natural stratification and planet life, the surface soil layer is more permeable compared to the subsurface layer.

scales. The primary contributions of this study are the following: (1) introduce a simple and accurate model for infiltration into layered soils; (2) precise physical and mathematical analyses of infiltration into two-layered soils; (3) estimations of the hydraulic conductivity and the wetting front matric head of each soil layer from a constant head infiltration into layered soils; and (4) an extension of the results to some practical applications. To achieve these goals, a new simple version of the Green–Ampt model for layered soils is proposed. To evaluate the proposed model, infiltration tests under a constant head were conducted on six different two-layered laboratory soil columns. Furthermore, the HYDRUS-1D numerical model which was based on Richards' equation is used to simulate infiltration into studied two-layered soil columns.

2. Proposed theoretical method

A soil profile under constant head infiltration consists of n layers with a thickness of L_i ($i = 1, 2, \ldots, n$). The hydraulic conductivity, the wetting front matric head and the wetness increment of each soil layer are K_i, Ψ_{fi} and $\Delta\theta_i$, respectively. Assuming that each soil layer is homogenous with a uniform initial water content; the layer interfaces are sharp; the available infiltration rate at the upper interface of each soil layer is greater than the infiltration rate capacity of the layer; and the water content behind the wetting front is uniform, the Green–Ampt approach is used to simulate the constant head infiltration into layered soils. When the wetting front advances through the 1st layer to a depth of L ($0 \leqslant L \leqslant L_1$), the infiltration rate dI/dt or the velocity of the water entering the soil can be obtained using Darcy's law. If the ground surface is the reference level, the total head at the ground surface and the total head at the wetting front are equal to H and $-L + \Psi_{f1}$, respectively, where H is the constant water depth above the soil surface (cm) [Fig. 2]; L is the thickness of wetting zone (cm); and Ψ_{f1} is the wetting front matric head of the 1st layer (cm). Applying Darcy's law between the ground surface and the wetting front, the infiltration rate can be expressed as:

$$\frac{dI}{dt} = -K_1 i = -K_1 \frac{(-L + \Psi_{f1}) - H}{L} = K_1 \frac{L + H - \Psi_{f1}}{L} \tag{1}$$

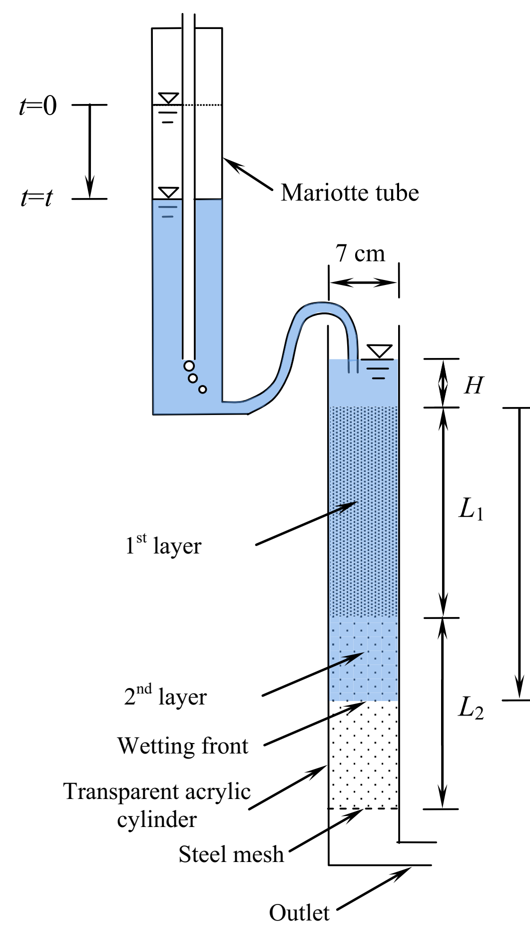

Fig. 2. Laboratory setup of constant head infiltration into a two-layered soil column.

where K_1 is the hydraulic conductivity of the 1st layer (cm/min); and i is the hydraulic gradient (cm/cm). During the infiltration process, L increases with time; therefore, based on Eq. (1), the hydraulic gradient and the infiltration rate must decrease with time. Because the accumulated amount of infiltrated water I is equal to $\Delta\theta_1 L$, Eq. (1) can be expressed in the following form:

$$\frac{dI}{dt} = K_1 \frac{I + \Delta\theta_1 (H - \Psi_{f1})}{I} \tag{2}$$

where $\Delta\theta_1$ is the wetness increment in the 1st layer. Integrating Eq. (2), the infiltration equation in the time range of $0 \leqslant t \leqslant t_1$ can be described as:

$$t = \frac{1}{K_1}\left[I - \Delta\theta_1 (H - \Psi_{f1}) \ln\left(\frac{I + \Delta\theta_1 (H - \Psi_{f1})}{\Delta\theta_1 (H - \Psi_{f1})}\right)\right] \tag{3}$$

where t_1 is the wetting front arrival time at the bottom of the 1st layer. When the wetting front advances to the ith layer ($i = 2,3,\ldots,n$), the infiltration rate equation can be expressed using Darcy's law:

$$\frac{dI}{dt} = \overline{K}\frac{L + H - \Psi_{fi}}{L} \tag{4}$$

where Ψ_{fi} is the wetting front matric head of the ith layer (cm); and $\overline{K}$ is the equivalent hydraulic conductivity from the ground surface to the wetting front (cm/min). $\overline{K}$ can be estimated by:

$$\overline{K} = \frac{L}{\frac{L_1}{K_1} + \frac{L_2}{K_2} + \cdots + \frac{L_{i-1}}{K_{i-1}} + \frac{L - \sum_{j=1}^{i-1} L_j}{K_i}} = \frac{L}{\sum_{j=1}^{i-1}\frac{L_j}{K_j} + \frac{L - \sum_{j=1}^{i-1} L_j}{K_i}} \tag{5}$$

Replacing $\overline{K}$ from Eq. (5) into Eq. (4) yields:

$$\frac{dI}{dt} = \frac{L + H - \Psi_{fi}}{\sum_{j=1}^{i-1}\frac{L_j}{K_j} + \frac{L - \sum_{j=1}^{i-1} L_j}{K_i}} \tag{6}$$

When the wetting front advances to the ith layer ($i = 2,3,\ldots,n$), the cumulative infiltration into the soil profile can be expressed using the mass conservation law:

$$I = I_{i-1} + \left(L - \sum_{j=1}^{i-1} L_j\right)\Delta\theta_i \tag{7}$$

where $\Delta\theta_i$ is the wetness increment in the ith layer; I_{i-1} is the stored infiltrated water from the ground surface to the bottom of the $(i-1)$th layer or the cumulative infiltration depth at $t = t_{i-1}$; and t_{i-1} is the wetting front arrival time at the bottom of the $(i-1)$th layer. Replacing L from Eq. (7) in Eq. (6), the infiltration rate equation is then:

$$\frac{dI}{dt} = \frac{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{\Delta\theta_i \sum_{j=1}^{i-1}\frac{L_j}{K_j} + \frac{I - I_{i-1}}{K_i}} \tag{8}$$

Eq. (8) can also be expressed in the following form:

$$dt = \left[\sum_{j=1}^{i-1}\frac{L_j}{K_j}\left(\frac{\Delta\theta_i}{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}\right) + \frac{1}{K_i}\left(1 - \frac{\Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}\right)\right] dI \tag{9}$$

The mathematical equation between I and t in the ith layer is obtained by integrating the left hand side of Eq. (9) from t_{i-1} to t and the right hand side from I_{i-1} to I:

$$t = t_{i-1} + \frac{I - I_{i-1}}{K_i} + \Delta\theta_i\left(\sum_{j=1}^{i-1}\frac{L_j}{K_j} - \frac{\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{K_i}\right) \times \ln\left(\frac{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{\Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}\right) \tag{10}$$

In most cases, the hydraulic parameters of the layers are unknown and must be estimated based on infiltration test data. Using the proposed method of Mohammadzadeh-Habili and Heidarpour (2011), K_1 and Ψ_{f1} can be estimated based on measured infiltration test data in the time range of $0 \leqslant t \leqslant t_1$. To estimate K_i and Ψ_{fi} ($i = 2,3,\ldots,n$) from infiltration test data, Eq. (10) is first expressed as:

$$I - I_{i-1} - \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)\ln\left(\frac{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{\Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}\right) = K_i\left[(t - t_{i-1}) - \Delta\theta_i\sum_{j=1}^{i-1}\frac{L_j}{K_j}\ln\left(\frac{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{\Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}\right)\right] \tag{11}$$

The right-hand side of Eq. (11) is a function of t and I, while the left-hand side is a function of I. Therefore, this equation can be expressed in the following form:

$$\mathrm{f}(I) = K_i \mathrm{f}(t, I) \tag{12}$$

where

$$\mathrm{f}(t,I) = (t - t_{i-1}) - \Delta\theta_i\sum_{j=1}^{i-1}\frac{L_j}{K_j}\ln\left(\frac{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{\Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}\right) \quad \text{and}$$
$$\mathrm{f}(I) = I - I_{i-1} - \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)\ln\left(\frac{I - I_{i-1} + \Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}{\Delta\theta_i\left(\sum_{j=1}^{i-1} L_j + H - \Psi_{fi}\right)}\right) \tag{13}$$

Using the direct proportionality between f(I) and f(t,I), a stepwise procedure presented in Appendix A is used to estimate the hydraulic conductivity and the wetting front matric head of each soil layer from the constant head infiltration test data.

When the physical and hydraulic parameters of a soil profile gradually change with depth, the soil profile can be divided into a large number of thin layers. Then, the hydraulic conductivity and the wetting front matric head of each soil layer can be estimated in a procedure similar to that of layered soils.

3. Laboratory setup and procedure

At the laboratory scale, the test conditions of infiltration into layered soils can be accurately selected and designed. Additionally, laboratory modeling is usually simpler and more economical than field experiments. It provides a better understanding of infiltration into layered soils that can be useful in practice. To model infiltration into two-layered soils in the lab, seven soils with different ranges of particle sizes were used. Soils A, B, C, D, E and F were uniform sandy soils consisting of fine-quartz spherical particles, and soil G was a clay soil. Because the primary portion of the low permeable heavy-textured soils usually consists of clay particles, soil G was used as a sample of these soils. Sandy soils A to F were also used as samples of permeable light-textured soils. Additionally, sandy soils with particles of different sizes provided different permeabilities. The physical properties of the studied

Table 1
Physical properties of the studied soils.

Soils	d (mm)	ρ_s (g/cm^3)
A	0.710–0.850	2.65
B	0.279–0.420	2.65
C	0.250–0.420	2.65
D	0.149–0.250	2.65
E	0.149–0.177	2.65
F	0.105–0.149	2.65
G	<0.074	2.25

soils are shown in Table 1, where d is the diameter range of the soil particles; and ρ_s is the specific density of the soil particles. For each soil, ρ_s was measured using the density bottle method. Soils A to F were free from organic materials; however, some organic material was present in soil G.

The laboratory infiltration tests on the two-layered soils were conducted in a transparent acrylic cylinder. The length and inner diameter of the cylinder were 60 cm and 7 cm, respectively. All soils were first dried in an oven at 105 °C for 24 h. For each test, the 2nd layer of soil was first placed in the acrylic cylinder and normally compacted with consecutive smooth hits of a rubber hammer on the cylinder body. Then, the 1st layer of soil was placed in the cylinder and was compacted in the same manner. Experimental observation indicated that firm compaction caused the soils to swell after wetting, and decreasing compaction causing soil subsidence. To simulate infiltration into the two-layered soils, a test setup was prepared, as shown in Fig. 2. A constant head of H was applied at the soil surface. The cumulative infiltration I into the soil column and thickness of the wetting zone L were recorded with time until the wetting front reached the bottom of the 2nd layer. Six different experiments were conducted: in tests 1 and 2, the 1st soil layer was more permeable; in the other tests, the 2nd soil layer was more permeable. Based on the analysis performed by Miller and Gardner (1962), the infiltration rate of tests 1 and 2 must be increased after the wetting front reached the interface of the two layers. Because the increase in the infiltration rate occurs over a short time period, a very high temporal resolution in the measurements of infiltration rate data was required (i.e., near $t = t_1$). In tests 1 and 2, to increase the reading accuracy of the infiltration rate data, the Mariotte tube surface area was set to be half of the soil column surface area; in the other tests, they were nearly equal. With the exception of the 2nd soil layer in tests 5 and 6, a sharp wetting front was observed in the other layers. The physical characteristics of the conducted tests are shown in Table 2, where Φ_1 is the porosity of the 1st layer, and Φ_2 is the porosity of the 2nd layer.

Table 2
Physical properties of the conducted tests on two-layered soil columns.

Test number	Soil of the 1st layer	Soil of the 2nd layer	H (cm)	L_1 (cm)	L_2 (cm)	Φ_1 (–)	Φ_2 (–)
1	D	G	3.70	26.40	26.80	0.476	0.541
2	E	G	2.70	30.65	22.75	0.450	0.482
3	E	C	4.20	26.05	26.10	0.456	0.412
4	G	E	4.00	27.55	22.95	0.511	0.459
5	F	B	4.00	29.80	22.40	0.447	0.417
6	F	A	3.35	27.45	26.20	0.442	0.417

The range of field porosities of the clay and fine sandy soils are respectively equal to 0.50–0.60 and 0.40–0.50 (Bouwer, 1978). As seen from Table 2, the porosity of each soil layer is within the range of field porosities. This result indicates good packing of the studied two-layered soil columns.

To simulate infiltration using HYDRUS-1D, the commonly used Van Genuchten's water retention curve model and Mualem's hydraulic conductivity model were used. Soil hydraulic parameters used for HYDRUS-1D simulations are presented in Table 3. where θ_r is the residual water content (–); θ_s is the saturated water content (–); K_s is the saturated hydraulic conductivity (cm/min); α (1/cm) and n are the empirical parameters; and subscripts 1 and 2 are the layer number. For each soil layer, l is taken equal to 0.5 and the initial water content θ_i is taken close to θ_r. As the studied soils were initially dry, setting $\theta_i \approx \theta_r$ make a small error in result of HYDRUS-1D.

4. Results and discussions

In the proposed model, it is assumed that the water content behind the wetting font is uniform. When the wetting front advances through the 1st layer ($0 \leqslant L \leqslant L_1$), the average wetness increment in the wetting zone can be estimated to be:

$$\Delta\theta = \frac{I}{L} \tag{14}$$

When the wetting front advances through the 2nd layer ($L_1 < L$), the average wetness increment in the wetting zone of the 2nd layer can be estimated to be:

$$\Delta\theta = \frac{I - I_1}{L - L_1} \tag{15}$$

where I_1 is the total infiltrated water to soil column at $t = t_1$; and t_1 is the wetting front arrival time at the bottom of the 1st layer. To determine the uniformity of the water content behind the wetting front and to measure the air content in the wetting zone, the wetness increment data of each layer were normalized based on the layer's porosity; these results are shown against the normalized wetting front thickness (Fig. 3). For the 1st layer (Fig. 3a), $\Delta\theta/\Phi_1$ is plotted against L/L_1, and for the 2nd layer (Fig. 3b), $\Delta\theta/\Phi_2$ is plotted against $(L - L_1)/L_2$. Due to the preferential flows presented in the 2nd layer during tests 5 and 6, the data of these two tests are not shown in Fig. 3b.

Fig. 3 shows that the water content behind the wetting front is approximately uniform, except in a thin soil layer on the soil surface in the 1st layer. For fine textured soils (i.e., clay), the normalized wetness increment behind the wetting front is closer to full saturation or zero air content in the wetting zone. Due to the additional attraction of the water and the small rate of wet front advancing in finer soils, the wetting zone becomes more saturated,

Table 3
Soil hydraulic parameters used for HYDRUS-1D simulations.

Test number	θ_{r1} (–)	θ_{r2} (–)	θ_{s1} (–)	θ_{s2} (–)	K_{s1} (cm/min)	K_{s2} (cm/min)	α_1 (1/cm)	α_2 (1/cm)	n_1 (–)	n_2 (–)
1	0.015	0.110	0.476	0.541	1.37	0.032	0.034	0.025	1.9	1.2
2	0.018	0.110	0.450	0.482	0.456	0.007	0.033	0.025	1.7	1.2
3	0.018	0.009	0.456	0.412	0.496	1.54	0.033	0.034	1.7	3.1
4	0.110	0.018	0.511	0.459	0.048	0.501	0.025	0.033	1.2	1.7
5	0.021	0.008	0.447	0.417	0.352	1.67	0.032	0.036	1.6	3.4
6	0.021	0.006	0.442	0.417	0.355	12.93	0.032	0.038	1.6	4.4

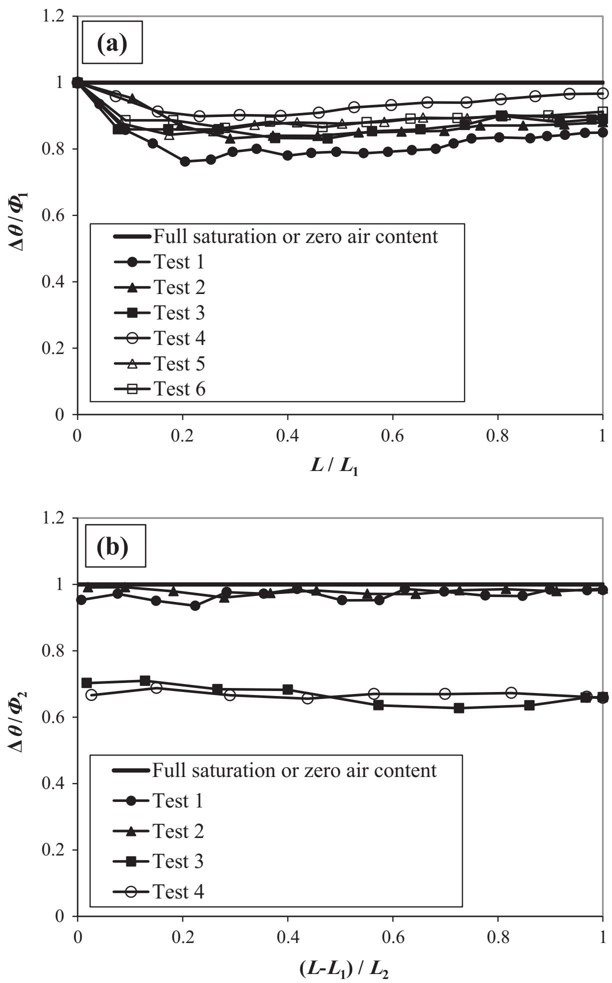

Fig. 3. Variation in the normalized wetness increment behind the wetting front with the normalized wetting front thickness (a) 1st layer [$\Delta\theta/\Phi_1$ against L/L_1] and (b) 2nd layer [$\Delta\theta/\Phi_2$ against $(L - L_1)/L_2$].

and the air content decreases. Additionally, due to the high permeability of the 2nd layers in tests 3 and 4, the rate of wet front advancing in the 2nd layer is high, while the available infiltrated water at the interface is small; this is due to the 1st layer's low permeability. The interaction between the large rate of wet front advancing and the low available infiltrated water at the interface increased the air content in the 2nd layers in tests 3 and 4. Additionally, based on the analysis performed by Chu and Mariño (2005), the air content in the 2nd soil layers in tests 3 and 4 must be large. Physically, gravity and wetting front matric head cause the water to flow in unsaturated soils; gravity causes a vertically downward water flow in the soil, while the wetting front matric head causes both vertical and horizontal flows. For soils with an approximately zero wetting front matric head, horizontal flows are minimal, and therefore, any infiltrated water tends to move vertically through the larger void tubes. If a zero wetting front matric head occurs with an insufficient amount of infiltrated water in the upper soil layer, little infiltrated water will move vertically through the larger void tubes, and the wetting front will become unstable. As shown by the analysis of Hill and Parlange (1972), instability in the wetting front can cause preferential flows in the 2nd layers of tests 5 and 6. When the wetting front of a ponded infiltration advances through the 2nd layer of a two-layered soil, Leconte and Brissette (2001) proposed that the soil behind the wetting front is entirely saturated; however, the data shown in Fig. 3 indicate that the wetted zone is not entirely saturated, and a considerable portion of the wetted zone is occupied by air.

To plot the infiltration rate curve, the wetness increment in both layers is required. Because the water content behind the wetting front is approximately uniform, $\Delta\theta_1$ and $\Delta\theta_2$ can be estimated using measured infiltration test data:

$$\Delta\theta_1 = \frac{I_1}{L_1}, \quad \Delta\theta_2 = \frac{I_2 - I_1}{L_2} \tag{16}$$

where I_2 is the total infiltrated water at $t = t_2$; and t_2 is the wetting front arrival time at the bottom of the 2nd layer. The air-filled volume fraction of the 1st and the 2nd layers Af_1 and Af_2, respectively, can also be estimated as:

$$\mathrm{Af}_1 = 1 - \frac{\Delta\theta_1}{\Phi_1}, \quad \mathrm{Af}_2 = 1 - \frac{\Delta\theta_2}{\Phi_2} \tag{17}$$

Using the mathematical definition of a limit, the shape of the infiltration rate curve when the wetting front arrives at the interface of the two layers can be described. The left and the right-hand limits of the infiltration rate function at $t = t_1$ should be obtained. For the left-hand limit of the infiltration rate function, the wetting front advances to the 1st layer and can be estimated as:

$$\lim \left(\frac{dI}{dt}\right)_{t \to t_1^-} = K_1 \frac{L_1 + H - \Psi_{f1}}{L_1} \tag{18}$$

For the right-hand limit of the infiltration rate function, the wetting front advances to the 2nd layer and can be estimated as:

$$\lim \left(\frac{dI}{dt}\right)_{t \to t_1^+} = K_1 \frac{L_1 + H - \Psi_{f2}}{L_1} \tag{19}$$

For two-layered soils, Ψ_{f1} is usually not equal to Ψ_{f2}; therefore:

$$\lim \left(\frac{dI}{dt}\right)_{t \to t_1^-} \neq \lim \left(\frac{dI}{dt}\right)_{t \to t_1^+} \tag{20}$$

Because the infiltration rate is a continuous function, the infiltration rate curve should continuously change from the left-hand limit to the right-hand limit. When $-\Psi_{f1} < -\Psi_{f2}$ (i.e., when the lower layer is usually finer), the right-hand limit is greater than the left-hand limit; therefore, the infiltration rate curve increases vertically from the left-hand limit to the right-hand limit. When $-\Psi_{f1} > -\Psi_{f2}$ (i.e., when the lower layer is usually coarser), the right-hand limit is smaller than the left-hand limit; therefore, infiltration rate curve is vertically reduced from the left-hand limit to the right-hand limit. In Fig. 4, the measured infiltration rate data of all of the tests are compared with the proposed theory and the HYDRUS-1D. For $t < t_1$, the infiltration rate curve of proposed theory is plotted using Eq. (2); for $t_1 < t$, it is plotted as: In tests 1 and 2, the 1st layer is more permeable, and at interface of the two layers, the available infiltration rate is larger than the infiltration rate capacity of the 2nd layer. Therefore, K_2 and Ψ_{f2} are estimated from the proposed method shown in Appendix A. The theoretical infiltration rate curve is also plotted using Eq. (8). As $-\Psi_{f1} < -\Psi_{f2}$, the right-hand limit of the infiltration rate function at $t = t_1$ is greater than the left-hand limit. Therefore, the infiltration rate curve increases vertically at $t = t_1$. For $t_1 < t$, a small value of K_2 acts as a limiting factor and causes a large reduction in the infiltration rate over a short time period. Then, the infiltration rate decreases smoothly with time. In test 1, due to the large difference in the wetting front matric heads of the two layers, an increase in the infiltration rate data is found. However, in test 2, due to the small difference in the wetting front matric heads of two layers, an increase in the infiltration rate data is not detected. Additionally, in tests 1 and 2, where a coarse soil layer is above a fine soil layer, the analysis proposed by Miller and Gardner

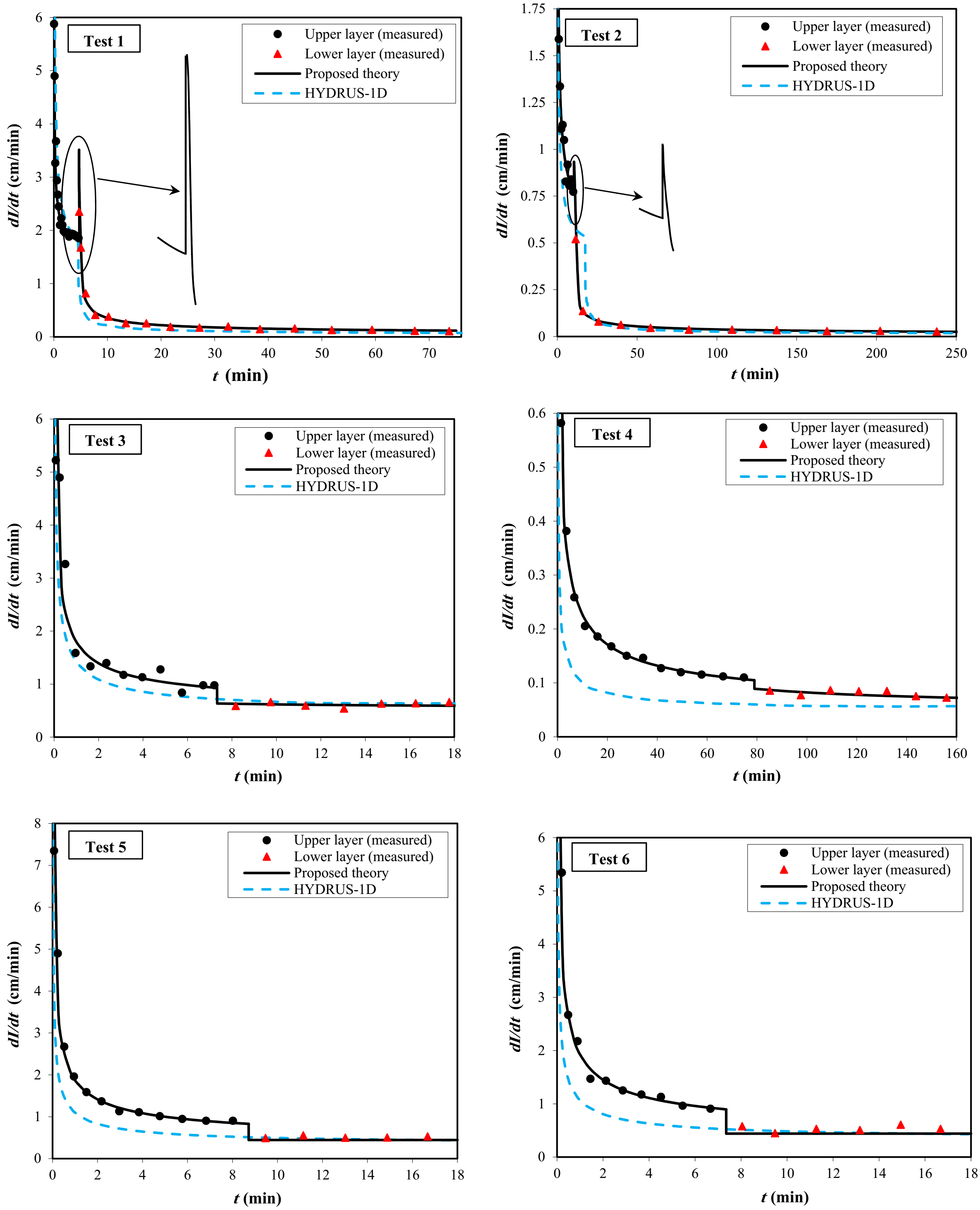

Fig. 4. The measured infiltration rate data into the studied two-layered soil columns compared to the proposed theory and the HYDRUS-1D (dI/dt versus t).

(1962) validates the vertical increase in the infiltration rate at $t = t_1$ as well as the significant reduction in the infiltration rate after $t = t_1$. The vertical increasing in infiltration rate of tests 1 and 2 is not predicted by HYDRUS-1D, while the rest of infiltration rate trend is well simulated. Physically, hydraulic conductivity describes the soil's resistance to water flow. When the 2nd soil layer is more permeable (i.e., tests 3–6), the rate of infiltrated water at the interface of the two layers is smaller than the potential infiltration rate capacity of the 2nd soil layer. Therefore, the resistance to water flow is low, and the infiltrated water can easily move through the 2nd soil layer. In this situation, the effect of the 2nd soil layer's hydraulic conductivity on the infiltration rate can be neglected, and K_2 cannot be estimated using infiltration test data. Conversely, Ψ_{f2} affects the infiltration and can be estimated using the infiltration test data. Neglecting the effect of K_2, the infiltration rate in the 2nd soil layer of tests 3 and 4 can be expressed as:

$$\frac{dI}{dt} = K_1 \frac{L + H - \Psi_{f2}}{L} \tag{21}$$

Combining Eq. (21) with Eq. (7) yields:

$$\frac{dI}{dt} = K_1 \frac{I - I_1 + \Delta\theta_2 (L_1 + H - \Psi_{f2})}{I - I_1 + \Delta\theta_2 L_1} \tag{22}$$

Eq. (22) can also be expressed as:

$$\frac{I - I_1 + \Delta\theta_2 L_1}{I - I_1 + \Delta\theta_2 (L_1 + H - \Psi_{f2})} dI = K_1 dt \tag{23}$$

Integrating the left-hand side of Eq. (23) from I_1 to I and the right-hand side from t_1 to t yields the cumulative infiltration equation:

$$I - I_1 - \Delta\theta_2 (H - \Psi_{f2}) \ln\left(\frac{I - I_1 + \Delta\theta_2 (L_1 + H - \Psi_{f2})}{\Delta\theta_2 (L_1 + H - \Psi_{f2})}\right) = K_1 (t - t_1) \tag{24}$$

The left-hand side of Eq. (24) is a function of I; therefore, this equation can be expressed in the following form:

$$\mathrm{f}(I) = K_1 (t - t_1) \tag{25}$$

Based on the direct proportionality between $\mathrm{f}(I)$ and $(t - t_1)$, the real value of Ψ_{f2} should produce a linear relationship between $\mathrm{f}(I)$ and $(t - t_1)$ with a slope of K_1. Using this characteristic, the wetting front matric heads of the 2nd soil layers in tests 3 and 4 were estimated. For $t_1 < t$, the infiltration rate curves in tests 3 and 4 are plotted using Eq. (22). Due to the preferential flows in the 2nd soil layers of tests 5 and 6, some portions of the 2nd soil layers' surface areas were not wetted. Additionally, the preferential flows in the 2nd soil layers indicate that $\Psi_{f2} \approx 0$; as a result, the pressure on the interface can be assumed to be atmospheric, and the infiltration rate equation can be expressed as:

$$\frac{dI}{dt} = K_1 \frac{L_1 + H}{L_1} = \text{constant} \tag{26}$$

For $t_1 < t$, the infiltration rate curve of tests 5 and 6 is plotted using Eq. (26). Both the measured data and theoretical curve indicate that the infiltration rate is constant. In tests 3 through 6, where a fine soil layer is above a coarse soil layer, the step reduction in the infiltration rate data at $t = t_1$ verified based on the analysis proposed by Latifi et al. (1994), while it is not predicted by HYDRUS-1D. Furthermore, a relative poor agreement is observed between measured infiltration rate data and HYDRUS-1D. For all of the tests, the good agreements between the measured infiltration rate data and the proposed theory indicate that the layers' hydraulic parameters are accurately estimated by the proposed method. Additionally, the good agreements indicate that the proposed theory can accurately simulate infiltration into two-layered soils. Using the proposed theory and measured infiltration rate data for ponded infiltration into two-layered soils, it has been shown that the trend of infiltration rate will tend to change when the wetting front reaches the interface of the two layers. When a coarse soil layer is above a fine soil layer, a larger $-\Psi_{f2}$ increases the hydraulic gradient when the wetting front first reaches the fine soil layer. Based on Darcy's law, an increase in the hydraulic gradient causes an increase in the water infiltration rate at the soil surface. As the wetting front advances into the finer soil layer, a smaller K_2 reduces the equivalent hydraulic conductivity and causes a reduction in the infiltration rate at the soil surface.

Table 4
Estimated hydraulic parameters of the conducted tests on two-layered soil columns.

Test number	$\Delta\theta_1$ (–)	$\Delta\theta_2$ (–)	K_1 (cm/min)	K_2 (cm/min)	Ψ_{f1} (cm)	Ψ_{f2} (cm)	Af_1 (–)	Af_2 (–)
1	0.404	0.532	1.582	0.038	−0.1	−28.2	0.151	0.017
2	0.396	0.476	0.499	0.007	−13.8	−23.0	0.119	0.013
3	0.406	0.272	0.546	–	−14.1	−0.1	0.109	0.341
4	0.494	0.298	0.051	–	−25.5	−16.9	0.033	0.321
5	0.401	–	0.392	–	−29.3	0.0	0.103	–
6	0.403	–	0.390	–	−32.2	0.0	0.088	–

When a fine soil layer is above a coarse soil layer, a smaller $-\Psi_{f2}$ reduces the hydraulic gradient when the wetting front first reaches the coarse soil layer. Based on Darcy's law, a reduction in the hydraulic gradient causes a reduction in the water infiltration rate at the soil surface.

For each test, the estimated values of $\Delta\theta_1$, $\Delta\theta_2$, K_1, K_2, Ψ_{f1}, Ψ_{f2}, Af_1 and Af_2 are shown in Table 4. In tests 3 and 4, the hydraulic conductivities of the 2nd soil layers are not estimated. Additionally, due to the preferential flows in 2nd soil layers of tests 5 and 6, $\Delta\theta_2$, K_2, and Af_2 are not estimated for these tests.

It follows from Table 4 that the estimated wetting front matric head for soil E in different conditions (e.g., when placed in the 1st layers of tests 2 and 3 with sufficient available water on the soil surface or in the 2nd layer of test 4 with the smallest available infiltrated water on the layer surface [less than 0.1 cm/min]) are nearly identical. The porosity and initial water content are two primary parameters that directly affect the wetting front matric head. Because the porosity and initial water content of soil E's layers are the same, the estimated values of the wetting front matric head for this system in its layers are nearly identical. The wetting front matric head of the 1st soil layer of test 1 and the wetting front matric head of the 2nd soil layer of test 3, which are finer than the 2nd soil layers of tests 5 and 6, are estimated to be nearly equal to zero. Therefore, the wetting front matric head of the 2nd soil layers of tests 5 and 6 must be equal to zero. This result shows that the preferential flows in the lower soil layer are an indicator of a zero wetting front matric head in the soil layer.

For layered soils in the field, when the layer interfaces are sharp, and each layer is homogenous with a uniform initial water content, the proposed theory can then be used to simulate ponded infiltration. For infiltration into two-layered soils, when a thick fine soil layer is above a coarse soil layer, the hydraulic gradient at $t = t_1$ tends to approach unity; therefore, the lower soil layer cannot affect the infiltration rate. As the thickness of a coarse soil layer above a fine soil layer increases, the hydraulic gradient at $t = t_1$ tends to approach unity, and therefore, the increase in infiltration rate at $t = t_1$ tends toward zero. However, the lower hydraulic conductivity of the fine soil layer reduces the equivalent hydraulic conductivity and produces a step reduction in the infiltration rate data. For a double ring infiltrometer test on homogenous and two-layered soils, the diameters of the rings can influence the infiltration; however, in two-layered soils, where a fine soil layer is above a coarse soil layer, the rings must be driven down to a depth below the interface of the two layers. If the rings are not driven into the coarse soil layer, the larger attraction of the upper soil layer for water absorption increases the lateral wetting below the edge of the rings. In this case, the infiltration rate increases compared to the one-dimensional wetting. Therefore, the layers' hydraulic parameters cannot accurately be estimated using the double ring infiltrometer test.

In addition to simulating infiltration into layered soils, some other applications for the results of this study include the following: (1) For layered soils with permeabilities that decrease with depth or a more permeable soil layer below a less permeable top soil layer, the infiltration rate and therefore percolation rate is significantly reduced after the wetting front arrives at each interface. In artificial recharge basins, reduction in percolation rate increases the water residence time in the basin and as a result water losses from evaporation will increase. Therefore, layered soils are not good systems for artificial recharge basins. To increase the recharge efficiency, the soil profile under the bed level of a recharge basin must be homogenous and highly permeable. (2) A coarse soil layer under a surface soil layer with a low permeability cannot effectively drain the surface soil layer. In this situation, a drainage system should be placed on the surface soil layer. (3) To cultivate agricultural crops with a ponding irrigation system (e.g., rice), a low permeable heavy-textured topsoil above another layer that is either more or less permeable can be a good system. In this situation, the infiltration rate at the soil surface decreases after the wetting front arrives at the interface of the two layers. The reduction in the infiltration rate causes a decrease in deep percolation and increases in the irrigation efficiency.

5. Conclusions

To simulate infiltration into layered soils and to estimate the layers' hydraulic parameters based on infiltration test data, a new simple version of the Green–Ampt approach is proposed. To evaluate the proposed method and to investigate the effect of the layers' permeability on infiltration, constant head infiltration tests were conducted on six different two-layered soil columns. The results indicated that the air content in the wetting zone of the 2nd soil layer depends on the available infiltrated water at the interface of the two layers and the permeability of the 2nd soil layer. A larger permeability increases the rate of wet front advancing in the 2nd soil layer. If a higher rate of wet front advancing is combined with a small amount of infiltrated water at the interface of the two layers (e.g., when the lower soil layer is more permeable than the upper soil layer), the air content in the wetting zone of the 2nd soil layer will increase. When the lower soil layer is significantly more permeable with an approximately zero wetting front matric head, the preferential flows or instability in the wetting front will occur in the lower soil layer. In two-layered soils where the surface soil layer is less permeable, the hydraulic conductivity of the lower soil layer cannot be estimated based on infiltration test data. When the surface soil layer is more permeable, the hydraulic parameters of both layers can be estimated. Good agreement between the measured infiltration rate data and the proposed theory indicates that the proposed theory can accurately simulate infiltration into two-layered soils. The proposed model in this study is simpler and more useful compared to sophisticated infiltration models that are computationally difficult to apply at large scales. The results of this study can also be useful in locating the best sites for artificial recharge basins, draining layered soils, cultivating agricultural crops on layered soils, and so on.

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Appendix A

The following stepwise procedure is proposed to estimate the hydraulic conductivity and the wetting front matric head of each layer using the constant head infiltration test data:

1. For $0 \leqslant t \leqslant t_1$, estimate K_1 and Ψ_{f1} based on the proposed method by Mohammadzadeh-Habili and Heidarpour (2011) for homogeneous soils.
2. Select a reasonable initial value for Ψ_{f2} based on the soil texture and the initial water content of the 2nd layer.
3. Using the measured infiltration test data for $t_1 \leqslant t \leqslant t_2$ and the selected value for Ψ_{f2} in step 2, calculate $f(t, I)$ and $f(I)$ from Eq. (13), and display $f(I)$ against $f(t,I)$ on a Cartesian coordinate plane.
4. Plot a straight line with best highest fit from the coordinate origin through the data points.
5. If all of the data points fit along a straight line, then the selected value for Ψ_{f2} in step 3 is correct, and the slope of the fitted line is equal to K_2; otherwise, change the value of Ψ_{f2} and repeat steps 2–4 until all of the data points fit along a straight line.
6. Using a similar procedure to that used with the 2nd layer, estimate the hydraulic conductivity and the wetting front matric head of the other layers.

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